

## Module-2

# RANDOM VARIABLES

### Random experiment (or) Trial

Tossing coin – Random experiment

Outcome - event

Sample space- Total number of outcomes

Sample point- any one outcome

Finite sample space- the number of sample point in sample space is finite

### Random variables:

A real variable 'X' whose value is determined by the outcome of a random experiment is called a random variable.

A real valued function defined on S and taking values in  $R(-\infty, +\infty)$  is called a one-dimensional random variables. If the values are ordered pairs of real numbers then the function is said to be two-dimensional random variables.

### Discrete random variable:

A random variable which takes a countable number of values is called a discrete random variable.

### Continuous random variable:

A random variable x is said to be continuous if it can take countably infinite values in an interval.

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### Probability mass function

If  $X$  is a discrete random variable takes countable number of values  $X_1, X_2, \dots$ , We associate a number  $P_i = P(X = x_i) = P(x_i)$ , called the probability mass function of  $X$ . The function  $P(x_i)$  satisfies the following conditions

$$(i) \quad P(x_i) \geq 0, \quad \forall \quad i = 1, 2, \dots, \infty$$

$$(ii) \quad \sum_{i=1}^{\infty} P(x_i) = 1$$

### Probability density function:

Consider a small interval  $(x, x + dx)$  of length  $dx$ . The function  $f(x)dx$  represents the probability that ' $X$ ' falls in the interval  $(x, x + dx)$ .

$$\text{ie., } P(x \leq X \leq x + dx) = f(x)dx$$

The probability function of a continuous random variable ' $X$ ' is called as probability density function and it satisfies the following conditions

$$(i) \quad f(x) \geq 0, \quad \forall x$$

$$(ii) \quad \int_{-\infty}^{+\infty} f(x)dx = 1$$

1. Verify whether the following is a probability distribution

| $X$    | 1    | 2    | 3    | 4    | 5    |
|--------|------|------|------|------|------|
| $P(x)$ | 1/10 | 2/10 | 4/10 | 2/10 | 1/10 |

$$(i) \quad P(x) \geq 0, \quad \forall x$$

$$(ii) \quad \sum_x P(x) = 1/10 + 2/10 + 4/10 + 2/10 + 1/10 = 1$$

Since both the conditions are satisfied, the given distribution is a probability distribution.

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2. Verify whether the following is a probability density function

$$f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(i)  $f(x) > 0, \forall x \in (0,1)$

(ii) Consider  $\int_0^1 f(x) dx = \int_0^1 3x^2 dx = \left(\frac{3x^3}{3}\right)_0^1 = 1$

$f(x)$  is a probability density function. **Dr.K.KAVITHA**